

Movie Theater Database

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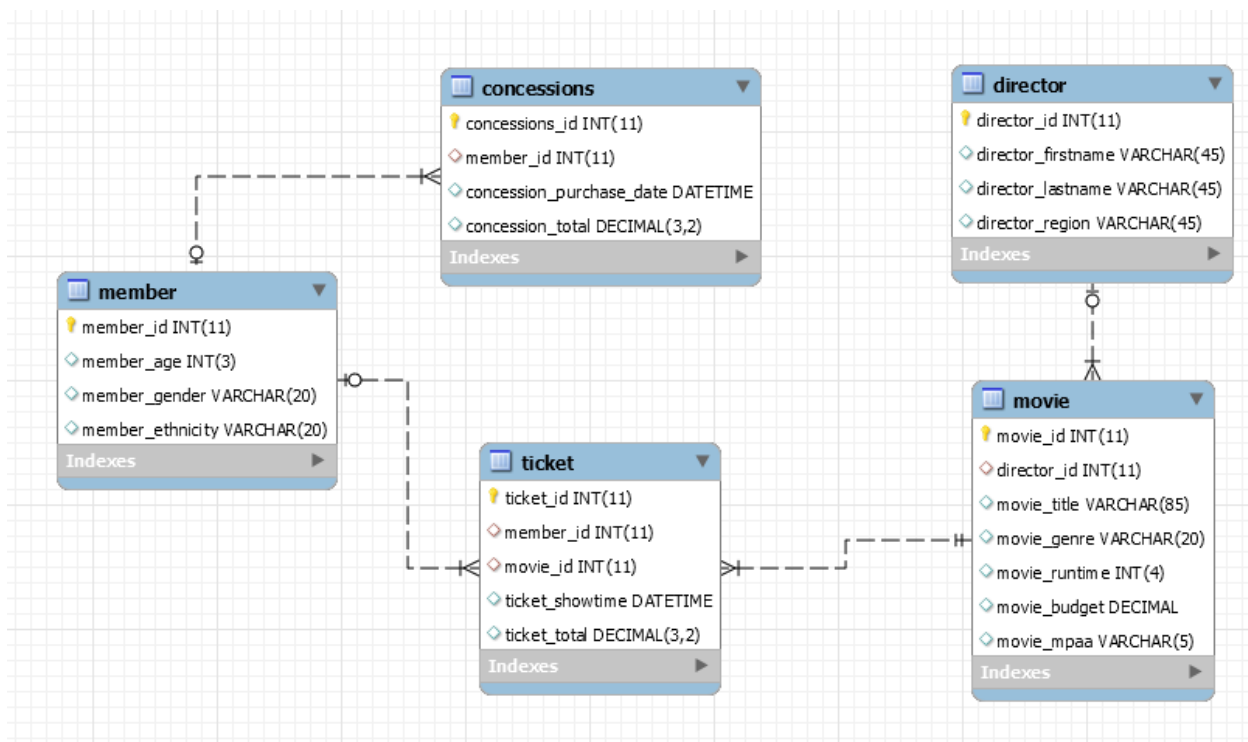
Summary

This database is designed to assist movie theaters in the observation of relevant statistics regarding ticket and concession purchase trends in relation to viewer demographics and film categories. It is assumed that the theater employing this database will have established a member system to track a member's ticket and concession purchases according to their member id.

In utilizing this information theaters can develop a deeper understanding of what films drive its community to attend the theater and make purchases. This information can encourage the theater to make more intelligent decisions about what movies to acquire and screen in the future, as well as how to best market the movies being screened.

For instance, a theater can examine what genre of film is driving the most concession sales on weeknights in September, or what age range has purchased the most tickets for small budget drama films during the summer. The idea is to use this information to make informed predictions about what specific combination of variables tend to drive the most sales.

ER Diagram



Normalization

The Movie Theater Database is normalized to the third degree. Normalization is the process of separating data into related tables. When we say whether a database is normal, typically we refer to the first three normal forms, although there are 7 total normal forms. In short, the first normal form states that a value must be stored as a scalar, and a table must not contain repeat columns. The second states that each non-key column must depend on the primary key, and the third states that each non-key column must depend only on the primary key.

In our database, each table has no repeating columns, indicating that it satisfies the first normal form. When it comes to the second normal form, every one of the non key columns in each table depends upon the primary key. As for the third normal form, each one of the non key columns depends only upon the primary key. Thus, our database design is normalized.